fn gamma_pow_eta_hi_from_x

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Proves soundness of `fn gamma_pow_eta_hi_from_x` in `mod.rs`.

1 Hoare Triple

Precondition

`eta` ≥ 0 and `x` is finite.

Pseudocode

```
1 def gamma_pow_eta_hi_from_x(eta: float, x: float):
2     gamma_hi = gamma_hi_from_x(x)  #  propagates helper errors
3     ensure_open_unit_interval(gamma_hi)  #  propagates helper errors
4     gamma_minus_one_hi = gamma_hi.inf_sub(1.0)  #  propagates helper errors
5     ln_gamma_hi = gamma_minus_one_hi.inf_ln_1p()  #  propagates helper errors
6     exponent_hi = eta.inf_mul(ln_gamma_hi)  #  propagates helper errors
7     return exponent_hi.inf_exp()  #  propagates helper errors
```

Postcondition

Theorem 1.1. For any inputs satisfying the precondition, `gamma_pow_eta_hi_from_x` either returns `Err(e)`, or returns `Ok(out)` where

$$\texttt{out} \geq (1 - e^{-\texttt{x}})^{\texttt{eta}}.$$

2 Proof

Assume the precondition and write $\gamma = 1 - e^{-x}$.

Lemma 2.1. `gamma_pow_eta_hi_from_x` only returns `Err(e)` if one of the fallible calls on lines 2, 3, 4, 5, 6, or 7 returns `Err(e)`.

Proof. These are exactly the fallible operations in the pseudocode. □

Lemma 2.2. If `gamma_pow_eta_hi_from_x` returns `Ok(out)`, then

$$\texttt{out} \geq (1 - e^{-\texttt{x}})^{\texttt{eta}}.$$

Proof. If line 3 succeeds, then `gamma_hi` lies in $(0, 1)$. By the proof of `gamma_hi_from_x`,

$$\texttt{gamma_hi} \geq \gamma.$$

Subtracting 1 preserves order, so

$$\texttt{gamma_minus_one_hi} \geq \gamma - 1.$$

Both values lie in $(-1, 0)$, therefore

$$\texttt{ln_gamma_hi} \geq \ln(\gamma).$$

Since $\texttt{eta} \geq 0$, upward-rounded multiplication gives

$$\texttt{exponent_hi} \geq \texttt{eta} \cdot \ln(\gamma).$$

Finally, upward-rounded exponentiation yields

$$\texttt{out} \geq e^{\texttt{eta}\ln(\gamma)} = \gamma^{\texttt{eta}} = (1 - e^{-x})^{\texttt{eta}}.$$

□

Proof of Theorem 1.1. Holds by Lemma 2.1 and Lemma 2.2. □